

Humanoid Tripod Robo-Receptionist– User Manual

Authors:

1. Tahir Abbas (**ID: 2020-EE-629**)
(B.Sc. Electrical Engineering)
University of Engineering and Technology (UET), Lahore
2. Zubair Usman (**ID: 2020-EE-630**)
(B.Sc. Electrical Engineering)
University of Engineering and Technology (UET), Lahore
3. Shahroz Khalid (**ID: 2020-EE-621**)
(B.Sc. Electrical Engineering)
University of Engineering and Technology (UET), Lahore

Submitted To:

*Department of Electrical Engineering
University of Engineering and Technology (UET), Lahore*

Date: May 2024

Supervisor:

Engr. Osama Bin Naeem

Lecturer, Department of Electrical Engineering

Purpose:

This user manual provides a detailed guide for setting up, operating, and maintaining the Humanoid Receptionist Robot, designed to serve as a stationary assistant capable of recognizing faces, providing directional guidance, and interacting with users via voice commands and a graphical user interface (GUI).

Table of Contents

Introduction	1
1.1 Purpose and Overview	1
1.2 Key Features.....	1
2. System Overview.....	1
2.1 Functional Requirements.....	1
2.2 System Architecture.....	2
2.3 System Components	2
2.4 Key Technologies Used	3
3. Getting Started	3
3.1 Prerequisites	3
3.2 Installation Instructions	4
3.3 Transfer the Project Files:	5
3.3 System Boot-Up	6
4. System Operation	7
4.1 User Interface Overview	7
4.2 Operational Modes	8
4.3 Query Response and Feedback.....	8
4.4 Voice Output System	8
5. Troubleshooting.....	8
5.1 Common Issues and Solutions	8
6. Safety Guidelines	9
6.1 General Safety.....	9
6.2 Electrical Safety.....	9
7. Appendix	10
8. Contact Information.....	11

Introduction

1.1 Purpose and Overview

The project aims to develop a Raspberry Pi-based Navigation System with speech and face recognition. It provides a user-friendly interface for guiding individuals, particularly those with visual impairments, through voice commands and visual feedback. The system integrates components like a Raspberry Pi Camera, Ultrasonic Sensor, Neck Joint Servo, and a touchscreen display. It uses speech recognition for command interpretation and face recognition for secure user identification, offering both indoor and outdoor navigation with real-time responses and auto-detection capabilities.

1.2 Key Features

- **Speech and Face Recognition:** Utilizes speech recognition technology to process user commands and face recognition for user identification.
 - **Indoor and Outdoor Navigation:** Capable of guiding users through indoor and outdoor spaces using pre-configured routes and dynamic querying.
 - **Voice and Visual Feedback:** The system provides audio feedback through speakers and visual feedback on a touchscreen display.
 - **Real-Time Query Responses:** Users can interact with the system to obtain real-time navigation instructions and queries, with voice outputs.
 - **User-Friendly Interface:** The touchscreen interface is intuitive, providing a guide and feedback section for seamless user interaction.
 - **Auxiliary Components:** Includes a neck joint servo for motion-based gestures, an ultrasonic sensor for distance measurement, and a microphone for voice commands.
 - **Data Storage:** Data related to navigation, user preferences, and feedback are stored in an SD card, allowing easy management and retrieval.
-

2. System Overview

The outlines of the components, functionalities, and architecture of System. The system utilizes speech and face recognition technologies to assist users in navigating their environment with ease.

2.1 Functional Requirements

- **Speech Commands:** Processes voice input for navigation and queries.
- **Face Recognition:** Ensures secure and personalized interactions.
- **Real-time Navigation:** Uses ultrasonic sensors for obstacle detection and movement assistance.

- **Feedback System:** Provides voice and touchscreen feedback for the user.
- **User Interface:** A simple, intuitive touchscreen interface for navigation and querying.

2.2 System Architecture

The system architecture includes the following key layers:

- **Input/Output Layer:** Consists of the microphone, Pi camera, touchscreen display, and speaker for user interaction.
- **Processing Layer:** Powered by the **Raspberry Pi**, responsible for speech recognition, face recognition, and navigation processing.
- **Auxiliary Components:** Includes the ultrasonic sensor for obstacle detection and the neck joint servo for guidance.
- **User Interface Layer:** Provides a GUI for navigation and querying.
- **Data Storage Layer:** Stores necessary data such as user preferences and system feedback in the **SD card**.

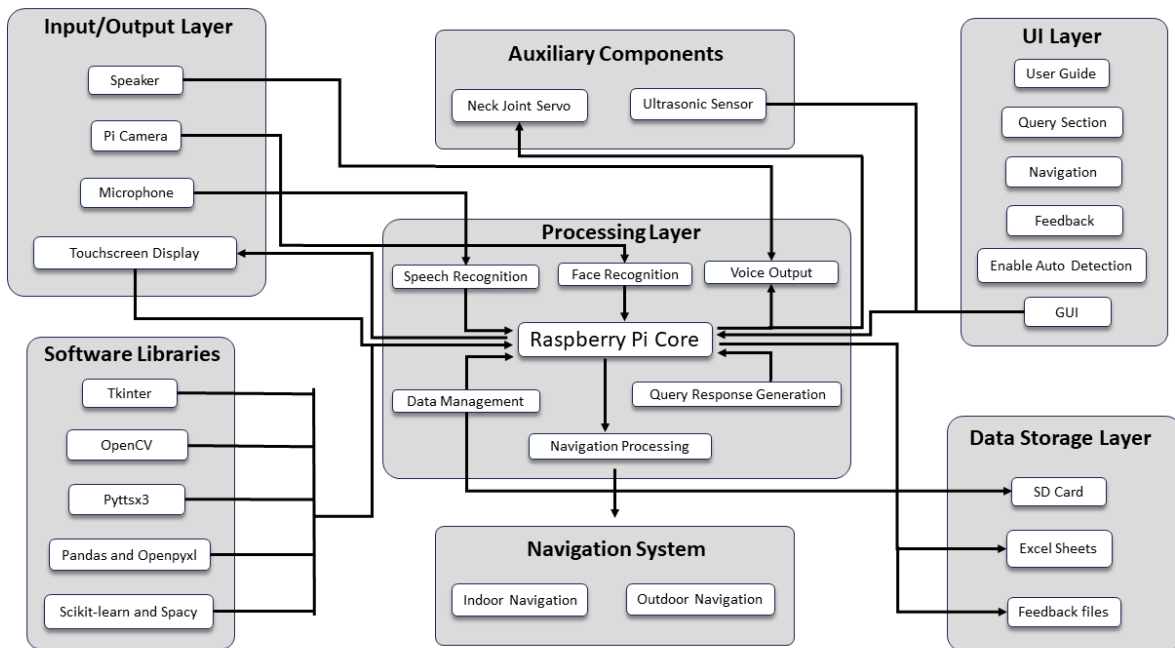


Fig 2.3.1 Block Diagram

2.3 System Components

Core components include:

- **Raspberry Pi 4:** Central processing unit.

- **Pi Camera (8MP):** Used for face recognition.
 - **Ultrasonic Sensor:** For obstacle detection and navigation.
 - **Neck Joint Servo (MG945):** Controls the device's physical movement.
 - **Speaker and Microphone:** For voice interaction.
 - **Touchscreen Display:** Visual interface for user input.
 - **SD Card (32GB):** Stores system data.
-

2.4 Key Technologies Used

Technologies and libraries used include:

- **Speech Recognition:** **SpeechRecognition** library.
 - **Face Recognition:** **OpenCV** and **Pytsx3**.
 - **Navigation:** Ultrasonic sensors for detecting obstacles.
 - **Raspberry Pi:** Runs all necessary software, including **Tkinter** (GUI) and **Pandas** (data management).
-

3. Getting Started

This section provides the essential steps for setting up and operating the Humanoid Receptionist Robot. It includes the hardware and software setup, as well as instructions on booting up the system.

3.1 Prerequisites

Before starting, ensure the following requirements are met:

- **Hardware:**
 - Raspberry Pi 4 with a power supply.
 - Pi Camera (8MP) for face recognition.
 - Ultrasonic Sensor for navigation.
 - Neck Joint Servo (MG945) for physical movements.
 - Touchscreen Display and Microphone for user interaction.
 - Speaker for audio feedback.
 - SD Card (32GB) with Raspberry Pi OS installed.
 - **Software:**
-

- Raspberry Pi OS (Raspberry Pi 4 compatible).
- Python 3.x and necessary Python libraries (listed in Appendix)

NOTE: In my system I have used python 3.11.7 but in raspberry pi it did not work for used libraries so in pi I have used python 3.9

3.2 Installation Instructions

Hardware Setup:

Connect Raspberry Pi Components:

- Attach the Pi Camera to the camera port of the Raspberry Pi.
- Connect the Ultrasonic Sensor to the GPO pins for distance measurement.
- Attach the Neck Joint Servo to the appropriate GPIO pin for movement.
- Connect the Touchscreen Display via HDMI and the micro-USB for power.
- Attach the Speaker and Microphone for voice input and output.
- Insert the SD Card into the Raspberry Pi.
- Connect the Raspberry Pi to a power supply to boot it up.

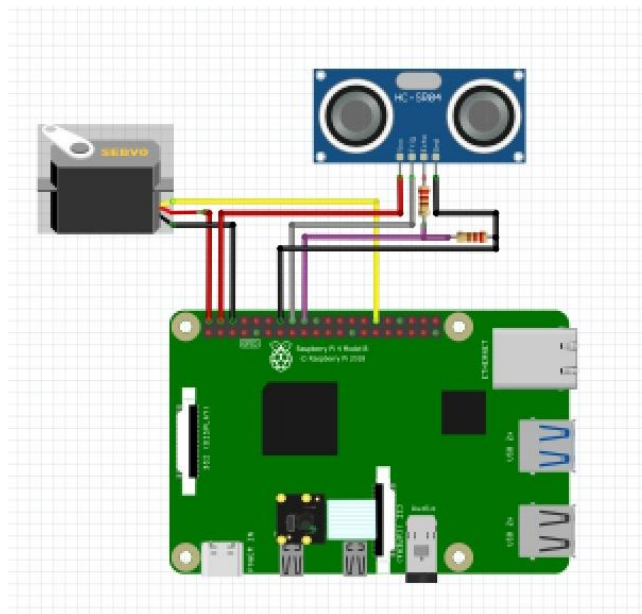


Fig 3.2.1 Configuration of sensor and servo

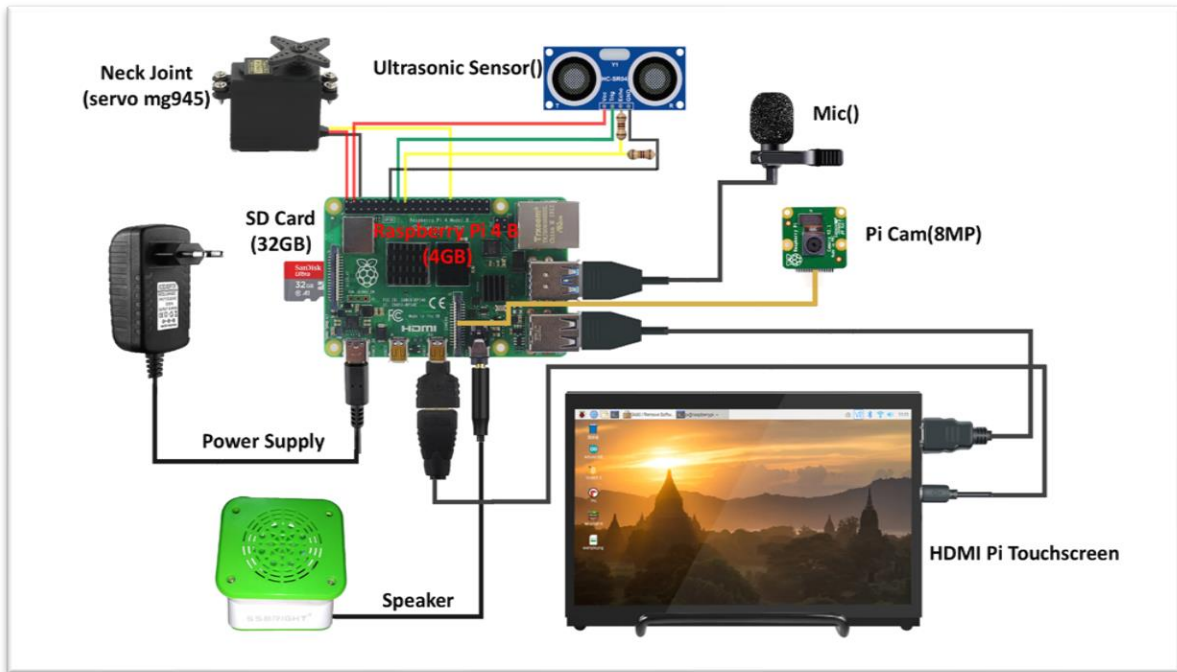


Fig 3.2.2 Circuit Diagram

Software Setup:

1. Install Raspberry Pi OS:
 - Ensure the Raspberry Pi OS is installed on the SD card. If not, follow the official Raspberry Pi OS installation guide.
["https://www.raspberrypi.com/documentation/computers/getting-started.html"](https://www.raspberrypi.com/documentation/computers/getting-started.html)
2. Install Required Python Libraries:
 - Open the terminal on the Raspberry Pi and run the following commands to install essential libraries:

```
sudo apt update
sudo apt install python3-pip
sudo pip3 install opencv-python
sudo pip3 install spacy
```

3.3 Transfer the Project Files:

Using FileZilla:

- Ensure that **FileZilla** is installed on both your computer and Raspberry Pi.
- Open FileZilla and connect to your Raspberry Pi using the following details:

1. **Host:** Raspberry Pi IP address (e.g., 192.168.1.55)
2. **Username:** pi
3. **Password:** raspberry
4. **Port:** 22

Transfer Files:

- Once connected, you will see your Raspberry Pi's file structure on the right side and your local system's files on the left side.
- Navigate to the directory where your project files are stored on your local system.
- Select the files and drag them to the desired directory on your Raspberry Pi (e.g., /home/pi/).

Navigate to the Project Directory:

- Once the files are transferred, open a terminal on the Raspberry Pi and navigate to the directory where you transferred the files:

```
cd /home/pi/your_project_folder
```

For detailed instructions on using FileZilla to transfer files to Raspberry Pi, refer to the official guide:

[“https://randomnerdtutorials.com/transfer-files-raspberry-pi-filezilla-ftp-windows/”](https://randomnerdtutorials.com/transfer-files-raspberry-pi-filezilla-ftp-windows/)

[“https://www.youtube.com/watch?v=gGGQJN20vSY”](https://www.youtube.com/watch?v=gGGQJN20vSY)

Configure Dependencies:

- Ensure that the Pi Camera is enabled by running:

```
sudo raspi-config
```

Then, navigate to Interface Options > Camera and enable it.

Test the Setup:

- Run a test script to ensure all hardware components are working correctly. This can include a script to test the **camera feed** 📷 📷, microphone, speaker, and ultrasonic sensor.

[“https://github.com/NimaTorbati/Raspberry-pi-camera/blob/main/capture_manual.py”](https://github.com/NimaTorbati/Raspberry-pi-camera/blob/main/capture_manual.py)

3.3 System Boot-Up

Once all components are connected and software is installed, follow these steps to boot up the system:

1. Power on the Raspberry Pi by connecting it to the power supply.

2. Wait for the system to initialize. The Raspberry Pi OS will load, and the system will be ready for interaction.
3. Run the Main Application: Execute the Python script responsible for initiating the reception robot's functionalities:

```
python3 receptionist_robot.py
```

NOTE: in our case main file name is main.py and remember to run this on thoany editor in raspberry pi OS

4. Initial Configuration:
 - On the first boot, the system will prompt for face registration to set up the face recognition database. Follow the on-screen instructions to add faces.
 - The system will initialize the navigation and voice response settings, ensuring that the ultrasonic sensor and microphone are calibrated for smooth operation.

4. System Operation

This section covers the main operations of the Humanoid Receptionist Robot, focusing on user interactions and operational modes.

4.1 User Interface Overview

The touchscreen display provides an intuitive interface for users to:

- Face Recognition: Register and recognize faces.
- Speech Command: Ask queries and receive responses.
- Navigation: Access step-by-step directions.



Fig 4.1 GUI

4.2 Operational Modes

1. Speech Recognition Mode: Listens for voice commands (e.g., "Where is the cafeteria?") and provides appropriate responses.
2. Face Recognition Mode: Detects and recognizes faces, greeting known users and providing a general welcome to unknown users.
3. Navigation Mode: Uses ultrasonic sensors to guide users and provide directions.

4.3 Query Response and Feedback

The system uses speech output (via pyttsx3) to provide answers and guidance, and visual feedback through the touchscreen for navigation directions.

4.4 Voice Output System

- The pyttsx3 library handles voice output, ensuring clear, customizable speech for commands and responses.

5. Troubleshooting

This section provides guidelines for troubleshooting to ensure the system operates efficiently.

5.1 Common Issues and Solutions

Here are some common issues you may encounter with the system, along with their solutions:

1. Face Recognition Not Working:

- Cause: Poor lighting conditions, or the face is not in the camera's field of view.
- Solution: Ensure proper lighting and adjust the angle of the camera. You can also re-train the system with better images of the user's face.

NOTE: All files to retrain the system are given separately.

2. Voice Commands Not Recognized:

- Cause: Background noise or microphone malfunction.
- Solution: Check the microphone connection and ensure there is minimal background noise during voice input. Adjust the sensitivity in the speech recognition settings.

3. System Not Booting Up:

- Cause: Power issues (**You can use your mobile charger specs 5V, 2.4A or 3A**) or corrupted SD card.

- Solution: Ensure the Raspberry Pi is properly connected to a stable power source. If the SD card is corrupted, re-image the SD card with the latest Raspberry Pi OS.

4. Servo Motor Not Responding:

- Cause: Incorrect wiring or insufficient power supply.
 - Solution: Check the wiring and ensure the motor is connected to the correct GPIO pins. Verify the power supply is sufficient for all components.
-

6. Safety Guidelines

This section outlines essential safety measures to ensure the safe operation of the Humanoid Receptionist Robot.

6.1 General Safety

- Avoid Overloading: Ensure that all components (e.g., power supply, Raspberry Pi) are not overloaded. Follow the manufacturer's specifications for voltage and current requirements.
 - Environment: Place the robot in a stable, dry environment to prevent damage from moisture or excessive heat.
 - Handling: Handle components, especially the Pi Camera and ultrasonic sensor, with care to avoid physical damage.
-

6.2 Electrical Safety

- Power Supply: Use the recommended 5V 3A power supply for the Raspberry Pi to avoid power fluctuations that could damage the system.
- Wiring: Ensure that all wiring is properly insulated to prevent short circuits, especially when dealing with the servo motors and ultrasonic sensors.
- Power Off During Maintenance: Always power off the system before performing any maintenance or hardware modifications.

7. Appendix

Technical Specifications

Component	Details
Raspberry Pi 4	Model: Raspberry Pi 4 Model B, Processor: Quad-core ARM Cortex-A72, 1.5 GHz, RAM: 4 GB, Storage: 32 GB SD card
Pi Camera (8MP)	Camera Module V2, Resolution: 3280 x 2464 pixels, Interface: CSI (Camera Serial Interface)
Ultrasonic Sensor (HC-SR04)	Measurement Range: 2 cm to 4 m, Accuracy: 3mm, Operating Voltage: 5V
Neck Joint Servo (MG945)	Operating Voltage: 5V, Torque: 13 kg-cm, Operating Speed: 0.17 s/60°
Speaker	Audio Output: Stereo, Connection: 3.5mm audio jack or Bluetooth
Microphone	Type: USB Microphone, Sensitivity: -38 dB

Libraries

Library	Purpose
Tkinter	GUI library for creating the user interface.
Tkimage	For image handling and display in Tkinter.
OpenCV-python	Image processing and face recognition.
OpenCv-contrib-python	Contributed OpenCV modules for advanced image processing tasks.
Pillow	Python Imaging Library for image handling.
SpeechRecognition	For processing voice commands and speech recognition.
Pytsx3 (pytz)	Text-to-speech conversion library.
Pandas	Data manipulation and analysis.
Openxl	Reading and writing Excel files.
Scikit-learn	Machine learning library for classification, clustering, etc.
Spacy	Natural language processing (NLP) library.
Matplotlib	Library for creating static, animated, and interactive visualizations in Python.
Chatterbot	A conversational dialogue engine for building chatbots.
Chatterbot Corpus	Pre-packaged datasets for the Chatterbot library.
Pyarrow (optional)	Cross-language development library for memory-mapped file formats, optional dependency for some data processing tasks.

8. Contact Information

For any inquiries or support regarding the **Humanoid Tripod Robo-Receptionist**, please use the following contact details:

Primary Contact

- **Name:** Tahir Abbas
- **Email:** ee.tahirabbas@gmail.com
- **Phone:** + 92- 324 0654306

Team Members

1. **Name:** Zubair Usman
Email: zubairusman858@gmail.com
Phone: +92 308 3724570
2. **Name:** Shehroz Khalid
Email: shehroz.khalid.pro@gmail.com
Phone: +92 349 2281261

Project Supervisor

- **Name:** Engr. Osama Bin Naeem
- **Position:** Lecturer, Department of Electrical Engineering
- **Email:** osamabinnaeem@gmail.com
- **Phone:** +92 303 0319195